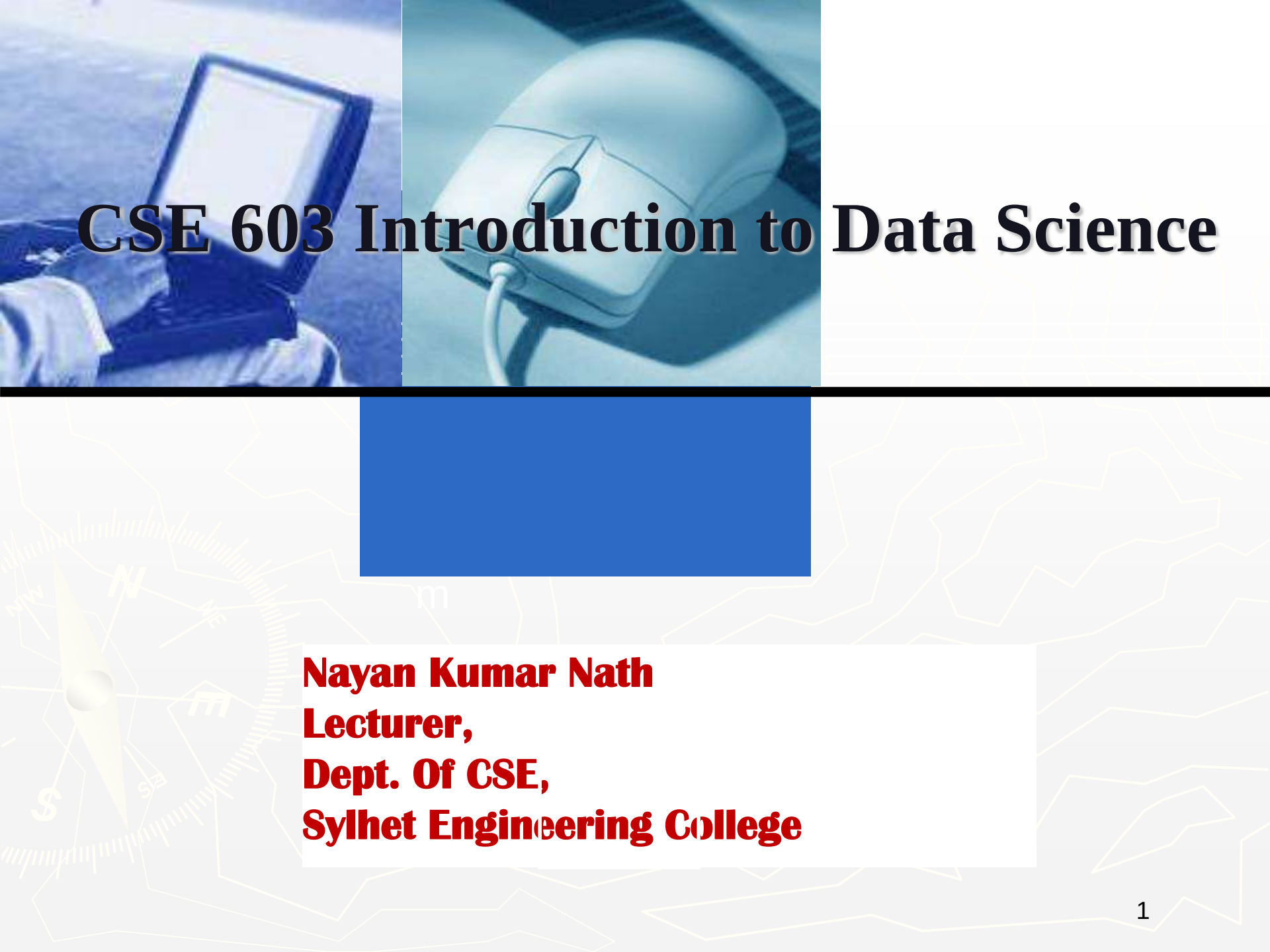
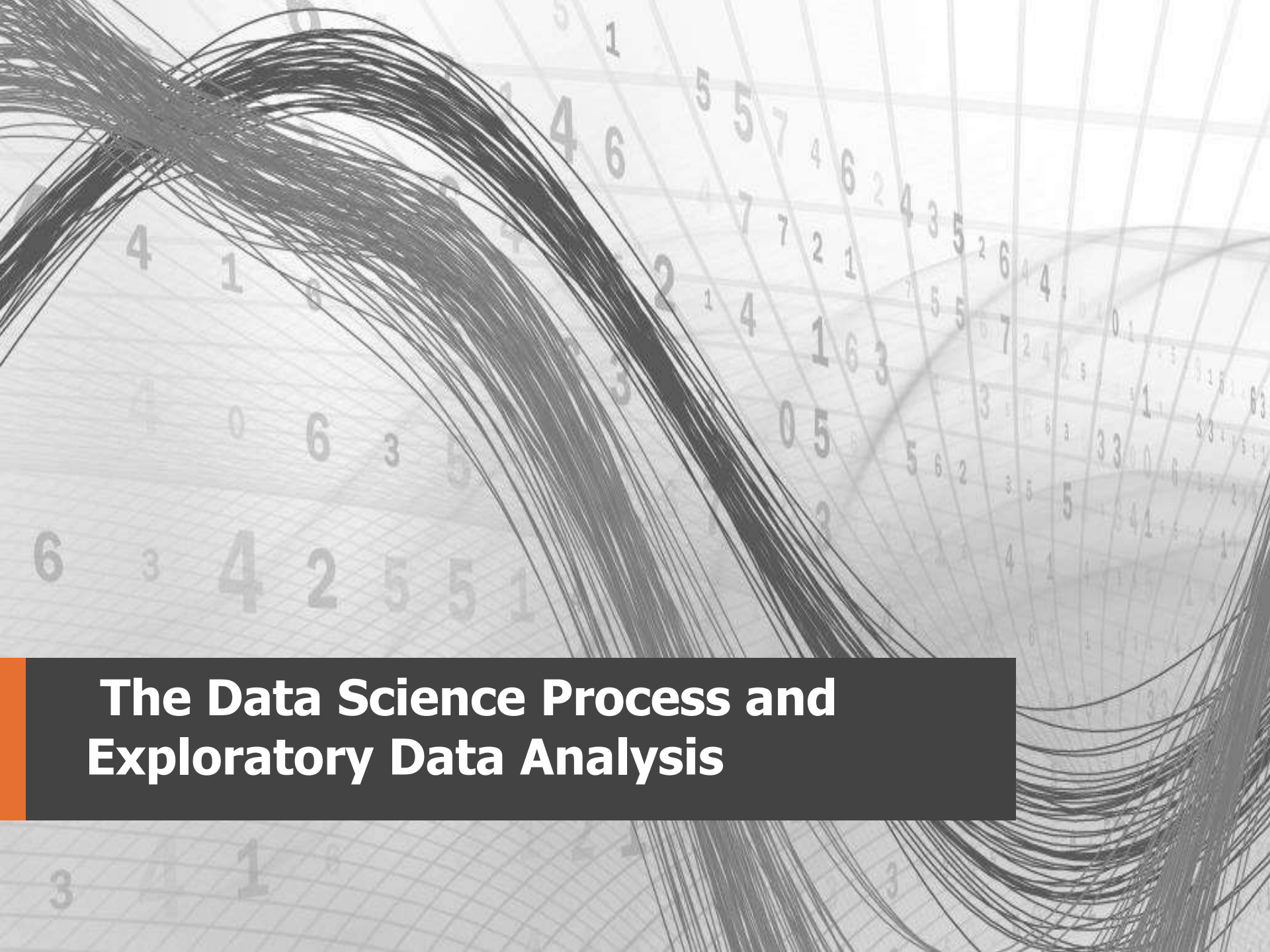




CSE 603 Introduction to Data Science

Nayan Kumar Nath
Lecturer,
Dept. Of CSE,
Sylhet Engineering College

The background of the slide is a complex, abstract composition. It features a dense network of thin, dark, curved lines that flow from the top left towards the bottom right, creating a sense of dynamic movement. Interspersed throughout this network are various numbers (0-9) in a light gray, sans-serif font. Some numbers are larger and more prominent, while others are smaller and more faded. The overall effect is one of data visualization and complexity.

The Data Science Process and Exploratory Data Analysis

Data Science Process

The methodical discovery of useful relationships and patterns in data is enabled by a set of iterative activities collectively known as the data science process.

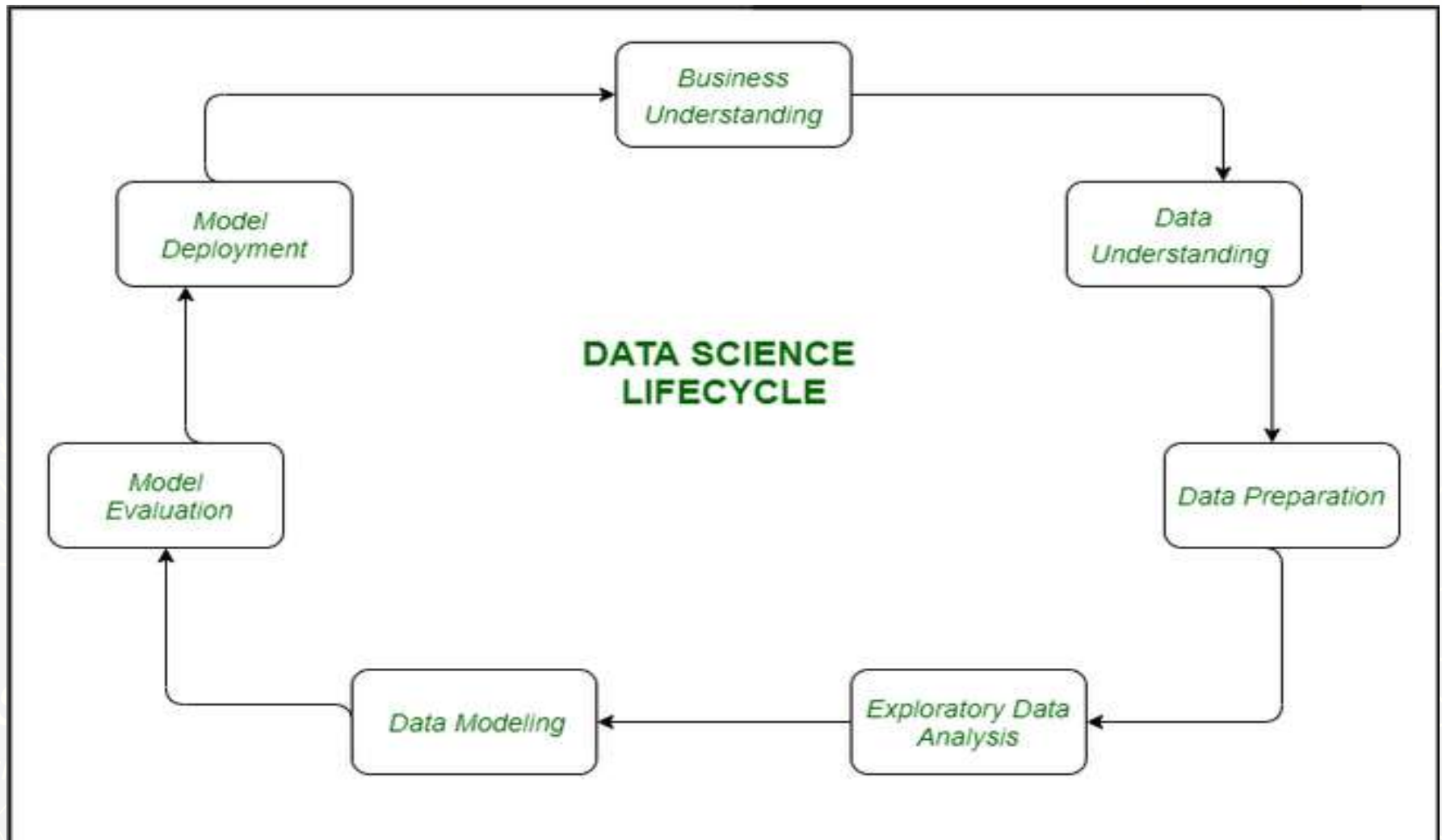
The standard data science process involves

- (1) understanding the problem,
- (2) preparing the data samples,
- (3) developing the model,
- (4) applying the model on a dataset to see how the model may work in the real world, and
- (5) deploying and maintaining the models

One of the most popular data science process frameworks is **Cross Industry Standard Process for Data Mining (CRISP-DM)**, which is an acronym for Cross Industry Standard Process for Data Mining. This framework was developed by a consortium of companies involved in data mining.

Data Science Process Life Cycle

There are some steps that are necessary for any of the tasks which are being done in the field of data science to derive any fruitful results from the data at hand.



Data Science Process

- 1. Business Understanding:** The complete cycle revolves around the enterprise goal. What will you resolve if you do not longer have a specific problem? It is extraordinarily essential to apprehend the commercial enterprise goal sincerely due to the fact that will be your ultimate aim of the analysis.
- 2. Data Understanding:** After enterprise understanding, the subsequent step is data understanding. This includes a series of all the reachable data.
- 3. Preparation of Data:** Next comes the data preparation stage. This consists of steps like choosing the applicable data, integrating the data by means of merging the data sets, cleaning it, treating the lacking values through either eliminating them or imputing them, treating inaccurate data through eliminating them, additionally test for outliers the use of box plots and cope with them.
- 4. Exploratory Data Analysis:** This step includes getting some concept about the answer and elements affecting it, earlier than constructing the real model. Distribution of data inside distinctive variables of a character is explored graphically the usage of bar-graphs, Relations between distinct aspects are captured via graphical representations like scatter plots and warmth maps.
- 5. Data Modeling:** Data modeling is the coronary heart of data analysis. A model takes the organized data as input and gives the preferred output. This step consists of selecting the suitable kind of model, whether the problem is a classification problem, or a regression problem or a clustering problem.

Data Science Process

6. Model Evaluation: Here the model is evaluated for checking if it is geared up to be deployed. The model is examined on an unseen data, evaluated on a cautiously thought out set of assessment metrics.

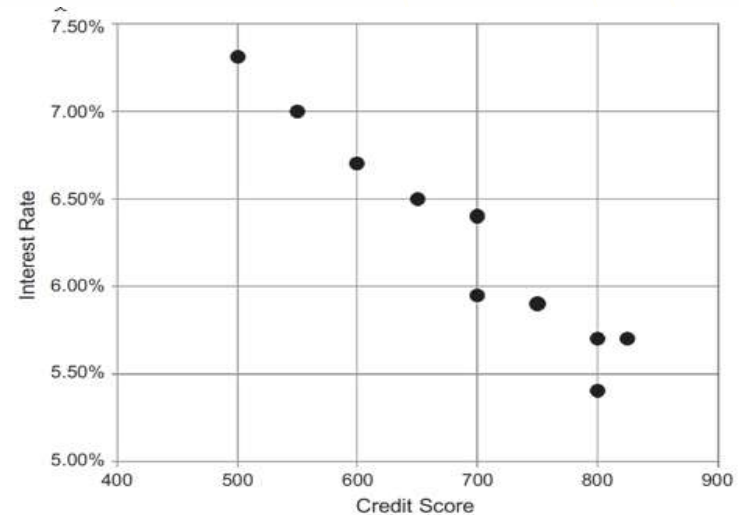
7. Model Deployment: The model after a rigorous assessment is at the end deployed in the preferred structure and channel. This is the last step in the data science life cycle. Each step in the data science life cycle defined above must be laboured upon carefully.



Data Science Process

Data Set – A dataset (example set) is a collection of data with a defined structure. example, a sample dataset of ten data points with three attributes has been put together: identifier, credit score, and interest rate

Borrower ID	Credit Score	Interest Rate
01	500	7.31%
02	600	6.70%
03	700	5.95%
04	700	6.40%
05	800	5.40%
06	800	5.70%
07	750	5.90%
08	550	7.00%
09	650	6.50%
10	825	5.70%



Data Preparation- Preparing the dataset to suit a data science task is the most time-consuming part of the process. It is extremely rare that datasets are available in the form A **Data Point:**(record, object or example) is a single instance in the dataset. Each row in Table 2.1 is a data point

Attribute: An attribute (feature, input, dimension, variable, or predictor) is a single property of the dataset

Label:A label (class label, output, prediction, target, or response) is the special attribute to be predicted based on all the input attributes. In Table 2.1, the interest rate is the output variable.

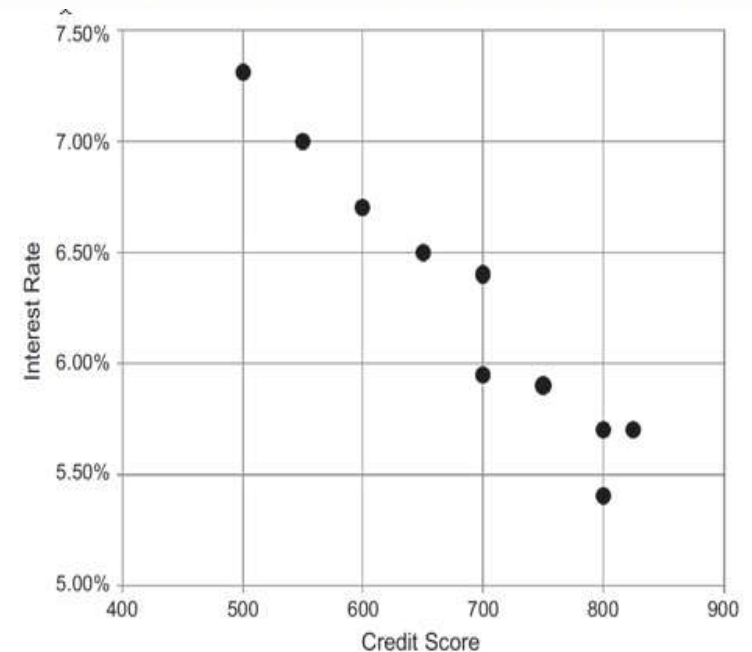
Identifiers: are special attributes that are used for locating or providing context to individual records. For example, common attributes like names, account numbers, and employee ID numbers are identifier attributes.

Data Science Process

Data Collection – After formulating any problem statement the main task is to calculate data that can help us in our analysis and manipulation. Sometimes data is collected by performing some kind of survey and there are times when it is done by performing

Table 2.1 Data Set

Borrower ID	Credit Score	Interest Rate
01	500	7.31%
02	600	6.70%
03	700	5.95%
04	700	6.40%
05	800	5.40%
06	800	5.70%
07	750	5.90%
08	550	7.00%
09	650	6.50%
10	825	5.70%



Data Preparation- Preparing the dataset to suit a data science task is the most time-consuming part of the process. It is extremely rare that datasets are available in the form required by the data science algorithms. This steps involves- Data Exploration, Data quality checking, Handling missing values, Data type conversion, Transformation, Outliers, Feature selection, Sampling

Data Science Process

- ▶ **Data Cleaning** – Most of the real-world data is not structured and requires cleaning and conversion into structured data before it can be used for any analysis or modeling.
- ▶ **Exploratory Data Analysis** – This is the step in which we try to find the hidden patterns in the data at hand. Also, we try to analyze different factors which affect the target variable and the extent to which it does so. How the independent features are related to each other and what can be done to achieve the desired results all these answers can be extracted from this process as well. This also gives us a direction in which we should work to get started with the modeling process.
- ▶ **Model Building** – Different types of machine learning algorithms as well as techniques have been developed which can easily identify complex patterns in the data which will be a very tedious task to be done by a human.

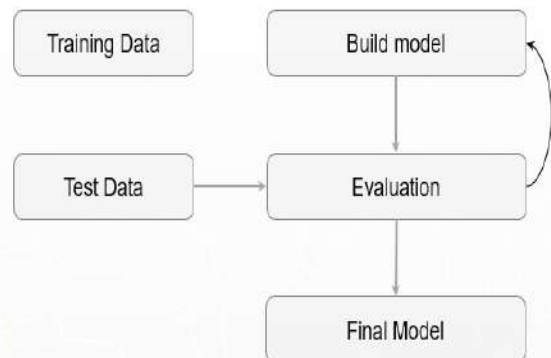


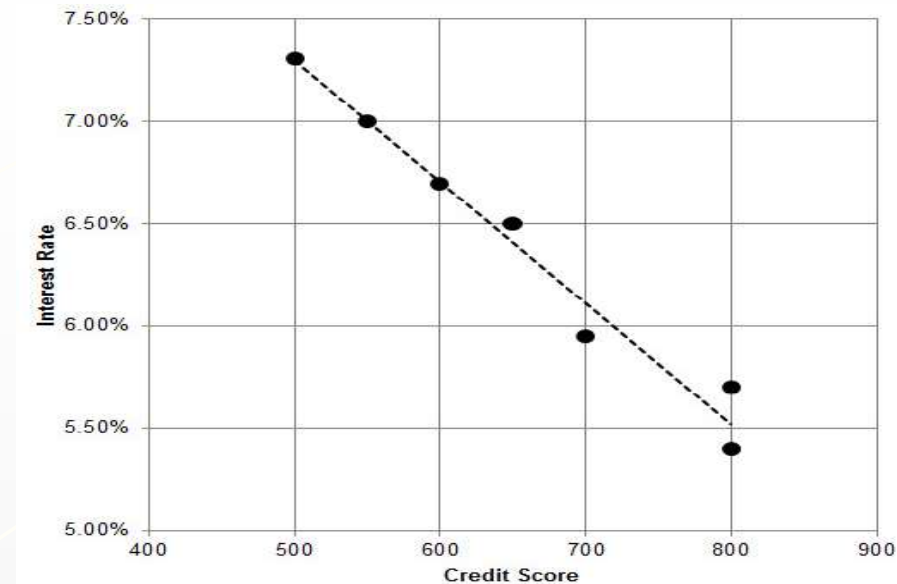
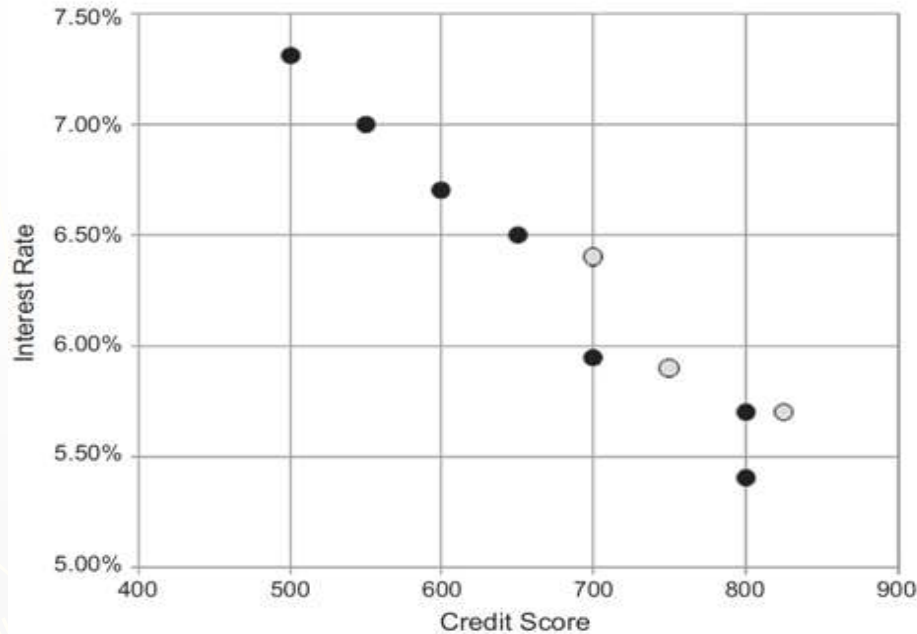
Table 2.3 Training Data Set

Borrower	Credit Score (X)	Interest Rate (Y)
01	500	7.31%
02	600	6.70%
03	700	5.95%
05	800	5.40%
06	800	5.70%
08	550	7.00%
09	650	6.50%

Table 2.4 Test Data Set

Borrower	Credit Score (X)	Interest Rate (Y)
04	700	6.40%
07	750	5.90%
10	825	5.70%

Data Science Process



- Training Data
- Testing Data

$$y = 0.1 + \frac{6}{100,000}x$$

Table 2.5 Evaluation of Test Data Set

Borrower	Credit Score (X)	Interest Rate (Y)	Model Predicted (Y)	Model Error
04	700	6.40%	6.11%	-0.29%
07	750	5.90%	5.81%	-0.09%
10	825	5.70%	5.37%	-0.33%

Usage of Data Science Process

- ▶ The Data Science Process is a systematic approach to solving data-related problems and consists of the following steps:
- ▶ **Problem Definition:** Clearly defining the problem and identifying the goal of the analysis.
- ▶ **Data Collection:** Gathering and acquiring data from various sources, including data cleaning and preparation.
- ▶ **Data Exploration:** Exploring the data to gain insights and identify trends, patterns, and relationships.
- ▶ **Data Modeling:** Building mathematical models and algorithms to solve problems and make predictions.
- ▶ **Evaluation:** Evaluating the model's performance and accuracy using appropriate metrics.
- ▶ **Deployment:** Deploying the model in a production environment to make predictions or automate decision-making processes.
- ▶ **Monitoring and Maintenance:** Monitoring the model's performance over time and making updates as needed to improve accuracy.

Tools for Data Science Process

As time has passed tools to perform different tasks in Data Science have evolved to a great extent. Different software like Matlab and Power BI, and programming Languages like Python and R Programming Language provides many utility features which help us to complete most of the most complex task within a very limited time and efficiently. Some of the tools which are very popular in this domain of Data Science are shown in the below image.



The background features a complex, abstract design. It consists of numerous thin, dark, curved lines that flow across the frame, creating a sense of motion and depth. Interspersed among these lines are various numbers (0-9) in a light gray font, some of which are slightly blurred or faded, giving the impression of data points or a digital landscape. The overall color palette is monochromatic, with shades of gray and black, accented by a small orange square on the left side of the title bar.

Data Exploration

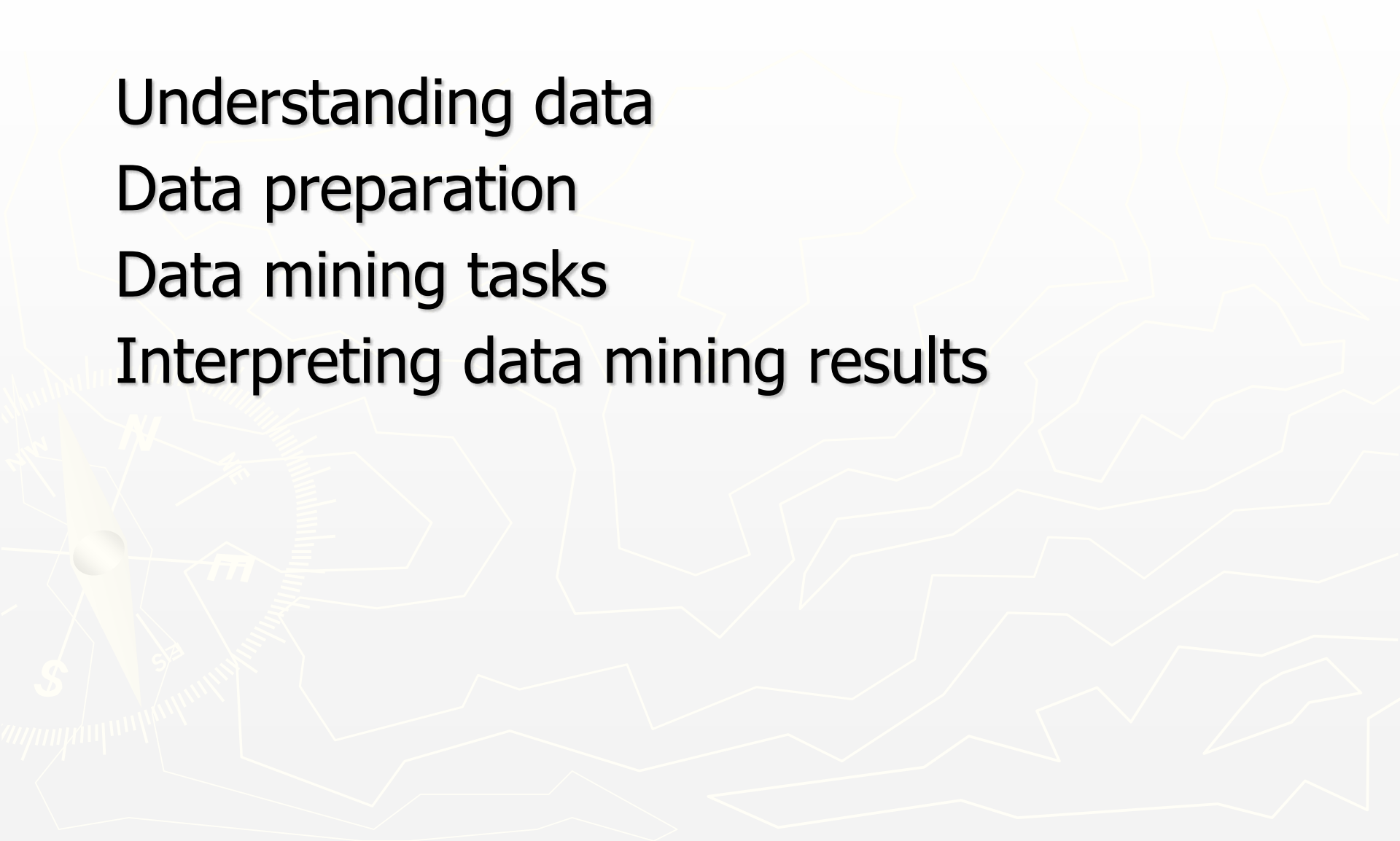
Objectives of Data Exploration

Understanding data

Data preparation

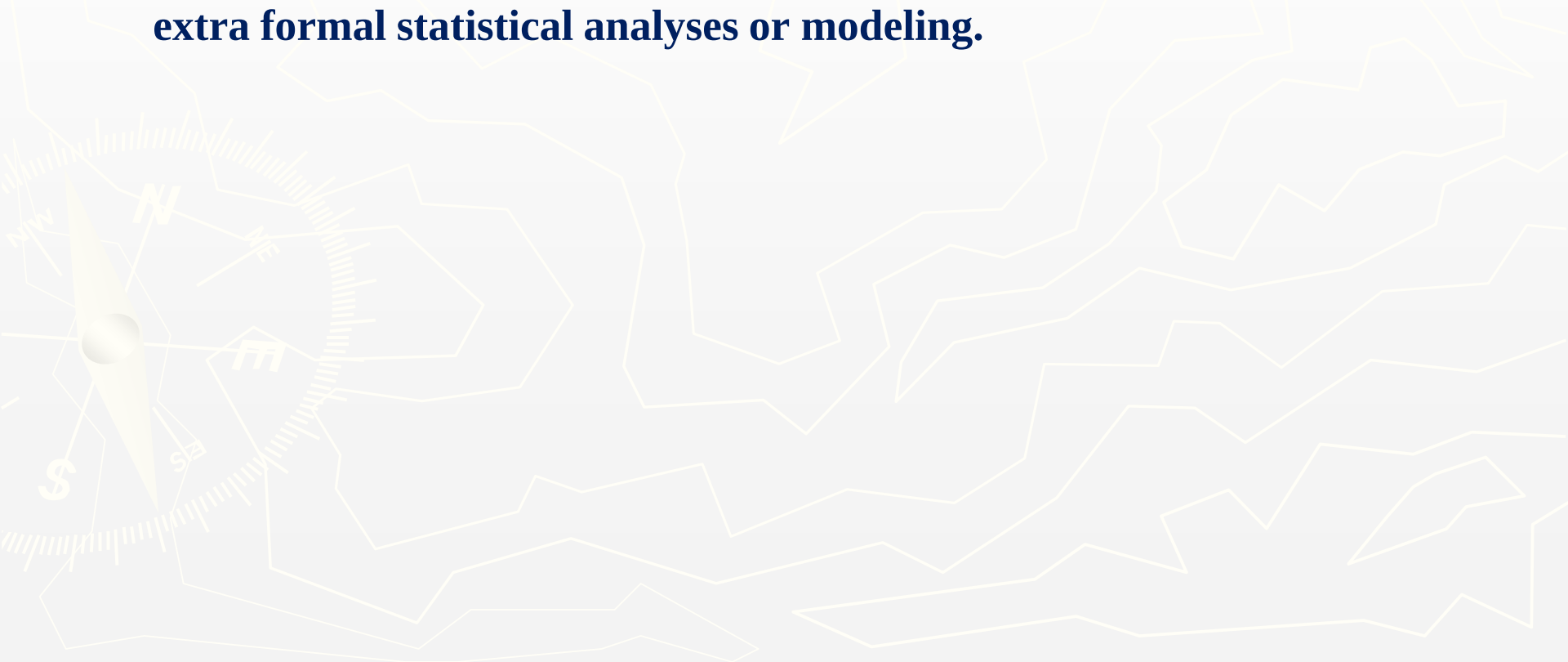
Data mining tasks

Interpreting data mining results



Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) refers to the method of studying and exploring record sets to apprehend their predominant traits, discover patterns, locate outliers, and identify relationships between variables. EDA is normally carried out as a preliminary step before undertaking extra formal statistical analyses or modeling.



The Foremost Goals of EDA

- ▶ **1. Data Cleaning:** EDA involves examining the information for errors, lacking values, and inconsistencies. It includes techniques including records imputation, managing missing statistics, and figuring out and getting rid of outliers.
- ▶ **2. Descriptive Statistics:** EDA utilizes precise records to recognize the important tendency, variability, and distribution of variables. Measures like suggest, median, mode, preferred deviation, range, and percentiles are usually used.
- 3. Data Visualization:** EDA employs visual techniques to represent the statistics graphically. Visualizations consisting of histograms, box plots, scatter plots, line plots, heatmaps, and bar charts assist in identifying styles, trends, and relationships within the facts.
- 4. Feature Engineering:** EDA allows for the exploration of various variables and their adjustments to create new functions or derive meaningful insights. Feature engineering can contain scaling, normalization, binning, encoding express variables, and creating interplay or derived variables.

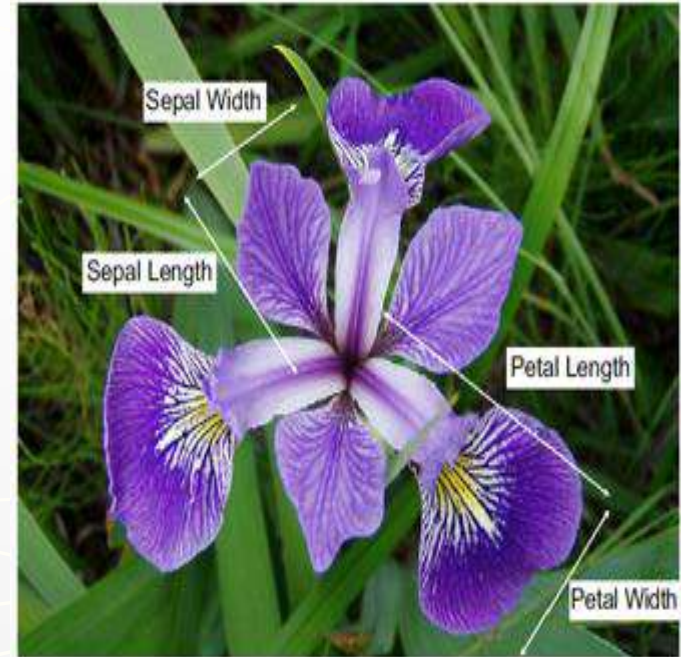
The Foremost Goals of EDA

- ▶ **5. Correlation and Relationships:** EDA allows discover relationships and dependencies between variables. Techniques such as correlation analysis, scatter plots, and pass-tabulations offer insights into the power and direction of relationships between variables.
- ▶ **6. Data Segmentation:** EDA can contain dividing the information into significant segments based totally on sure standards or traits. This segmentation allows advantage insights into unique subgroups inside the information and might cause extra focused analysis.
- ▶ **7. Hypothesis Generation:** EDA aids in generating hypotheses or studies questions based totally on the preliminary exploration of the data. It facilitates form the inspiration for in addition evaluation and model building.
- ▶ **8. Data Quality Assessment:** EDA permits for assessing the nice and reliability of the information. It involves checking for records integrity, consistency, and accuracy to make certain the information is suitable for analysis.

Data Sets

A dataset (example set) is a collection of data with a defined structure. This structure is also sometimes referred to as a “data frame”

The most popular datasets used to learn data science is probably the Iris dataset, introduced by Ronald Fisher, in his seminal work on discriminant analysis, "The use of multiple measurements in taxonomic problems" (Fisher, 1936). The Iris dataset contains 150 observations of three different species, *Iris setosa*, *Iris virginica*, and *I. versicolor*, with 50 observations each. Each observation consists of four attributes: sepal length, sepal width, petal length, and petal width.

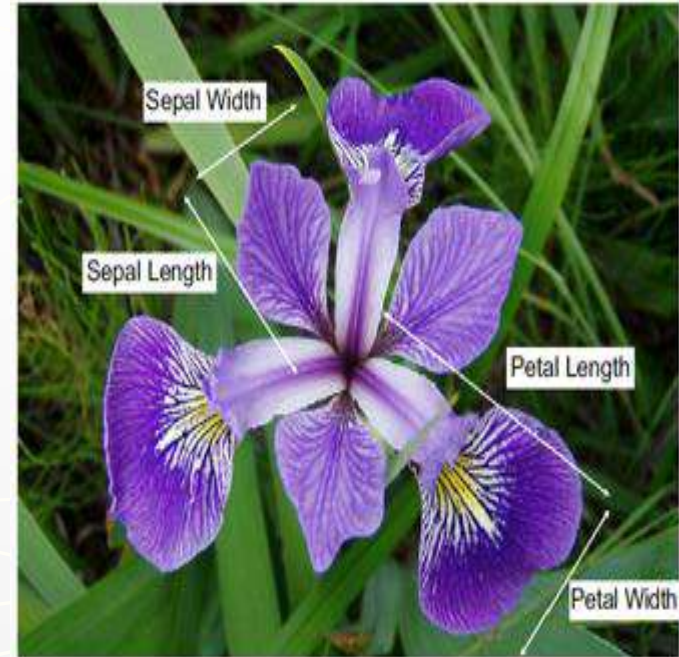


1http://commons.wikimedia.org/wiki/File:Iris_versicolor_3.jpg#mediaviewer/File:Iris_versicolor_3.jpg

Data Sets-Kaggle

Kaggle is a data science competition platform and online community for data scientists and machine learning practitioners under Google LLC.

Kaggle enables users to find and publish datasets, explore and build models in a web-based data science environment, work with other data scientists and machine learning engineers, and enter competitions to solve data science challenges. Kaggle was founded by [Anthony Goldbloom](#) in April 2010.^[2] [Jeremy Howard](#), one of the first Kaggle users, joined in November 2010 and served as the President and Chief Scientist.



1http://commons.wikimedia.org/wiki/File:Iris_versicolor_3.jpg#mediaviewer/File:Iris_versicolor_3.jpg

Data Sets-Kaggle

Several classic data sets have been used extensively in the statistical literature:

1) Iris flower data set – Multivariate data set introduced by Ronald Fisher(1936). Provided online by University of California-Irvine Machine Learning Repository.

2) MNIST database – Images of handwritten digits commonly used to test classification, clustering, and image processing algorithms

2) Categorical data analysis – Data sets used in the book, An Introduction to Categorical Data Analysis, provided online by UCLA Advanced Research Computing.

3) Robust statistics – Data sets used in Robust Regression and Outlier Detection (Rousseeuw and Leroy, 1968). Provided online at the University of Cologne.

4) Time series – Data used in Chatfield's book, The Analysis of Time Series, are provided on-line by StatLib.

5) Extreme values – Data used in the book, An Introduction to the Statistical Modeling of Extreme Values are a snapshot of the data as it was provided on-line by Stuart Coles, the book's author.

6) Bayesian Data Analysis – Data used in the book are provided on-line (archive link) by Andrew Gelman, one of the book's authors.

7) The Bupa liver data – Used in several papers in the machine learning(data mining) literature.

8) Anscombe's quartet – Small data set illustrating the importance of graphing the data to avoid statistical fallacies.

Descriptive Statistics - Univariate

Descriptive statistics refers to the study of the aggregate quantities of a dataset. In general, descriptive analysis covers the following characteristics of the sample or population dataset (Kubiak & Benbow, 2006).

Univariate data exploration denotes analysis of one attribute at a time.

Characteristics of the Data Set	Measurement Technique
Center of the data set	Mean, median, and mode
Spread of the data set	Range, variance, and standard deviation
Shape of the distribution of the data set	Symmetry, skewness, and kurtosis

Table 3.1 Iris Data Set and Descriptive Statistics (Fisher, 1936)

Observation	Sepal Length	Sepal Width	Petal Length	Petal Width
1	5.1	3.5	1.4	0.2
2	4.9	3.1	1.5	0.1
...	...	...	...	...
49	5	3.4	1.5	0.2
50	4.4	2.9	1.4	0.2
Statistics	Sepal Length	Sepal Width	Petal Length	Petal Width
Mean	5.006	3.418	1.464	0.244
Median	5.000	3.400	1.500	0.200
Mode	5.100	3.400	1.500	0.200
Range	1.500	2.100	0.900	0.500
Standard Deviation	0.352	0.381	0.174	0.107
Variance	0.124	0.145	0.030	0.011

Descriptive Statistics - Univariate



FIGURE 3.2

Descriptive statistics for the Iris dataset.

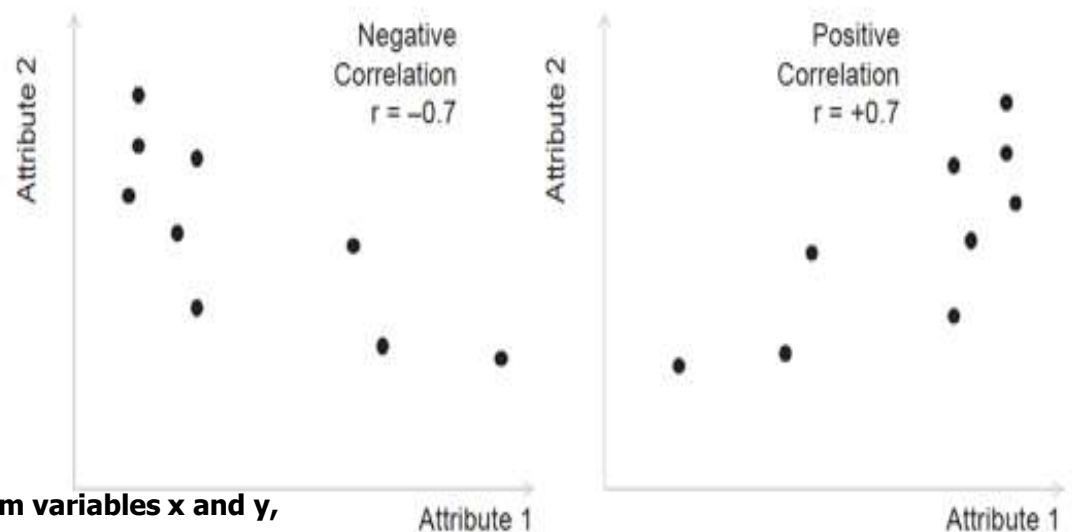
Descriptive Statistics - Multivariate

Multivariate exploration is the study of more than one attribute in the dataset simultaneously. This technique is critical to understanding the relationship between the attributes, which is central to data science methods. Similar to univariate explorations, we have to discuss the measure of central tendency and variance

Central datapoint: observation i : {sepal length, sepal width, petal length, petal width}

Correlation: Correlation measures the statistical relationship between two attributes, particularly dependence of one attribute on another attribute. When two attributes are highly correlated with each other, they both vary at the same rate with each other either in the same or in opposite directions

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$
$$= \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{N * S_x * S_y}$$



where s_x and s_y are the standard deviations of random variables x and y , respectively.

Descriptive Statistics - Multivariate

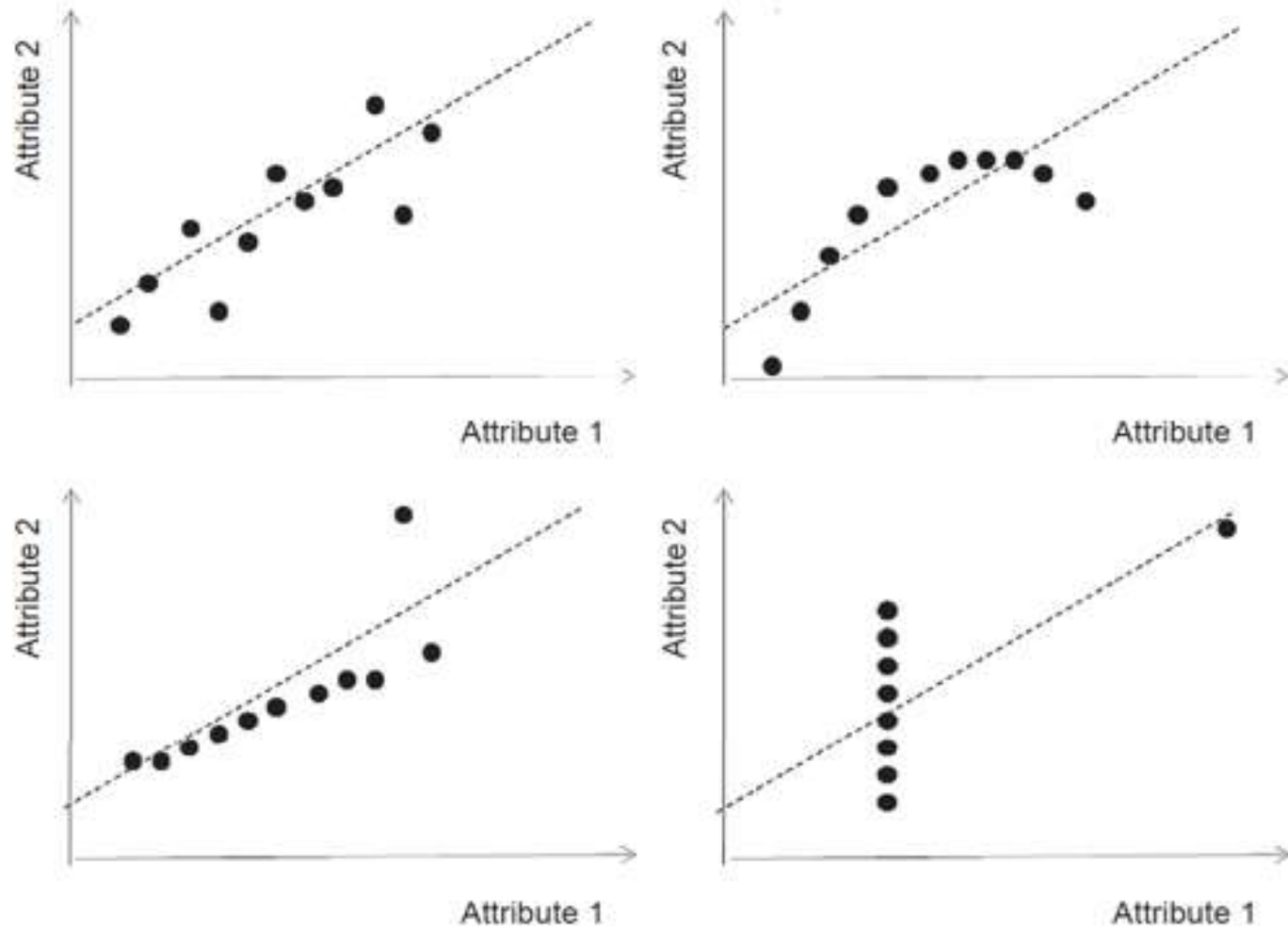


FIGURE 3.4

Anscombe's Quartet: descriptive statistics vs. visualization (Anscombe, F. J., 1973. Graphs in Statistical Analysis, *American Statistician* 27 (1), pp. 19–20.)

Data Visualization

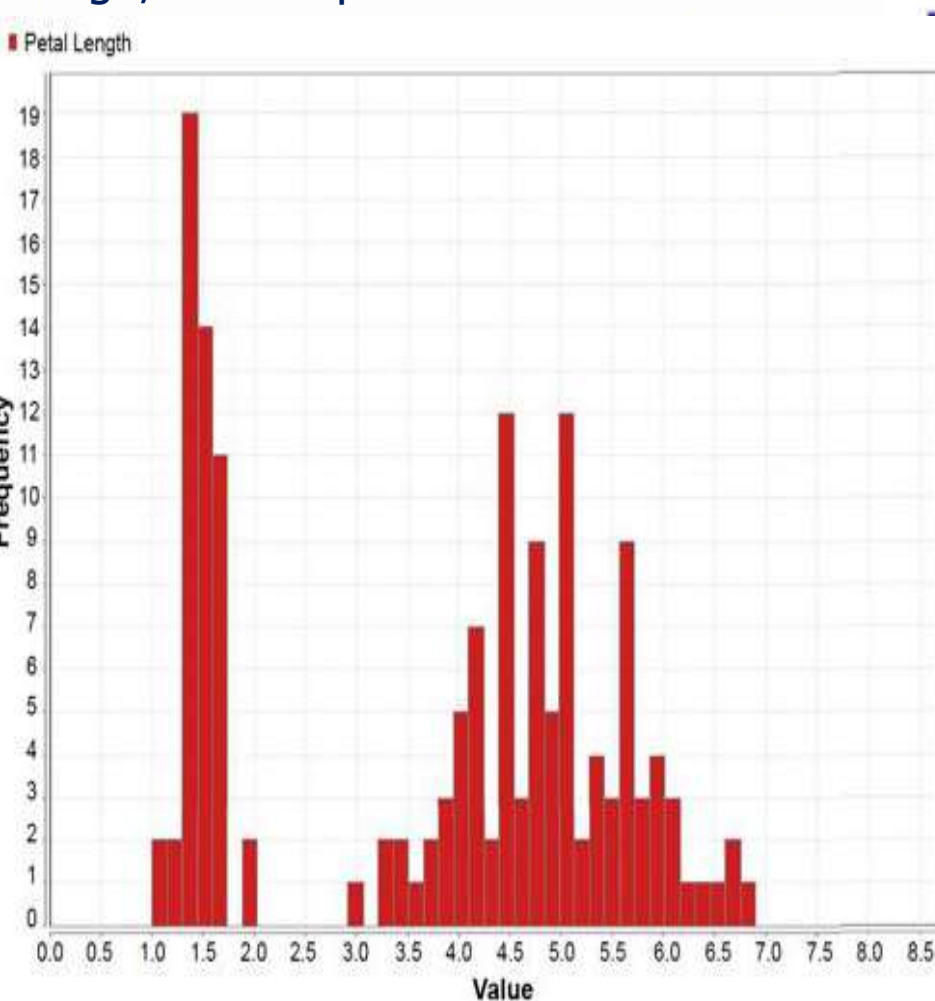
Data visualization encompasses the methods of expressing data in an abstract visual form. The visual representation of data provides easy comprehension of complex data with multiple attributes and their underlying relationships. The motivation for using data visualization includes:

1.Comprehension of dense information: A simple visual chart can easily include thousands of data points. By using visuals, the user can see the big picture, as well as longer term trends that are extremely difficult to interpret purely by expressing data in numbers

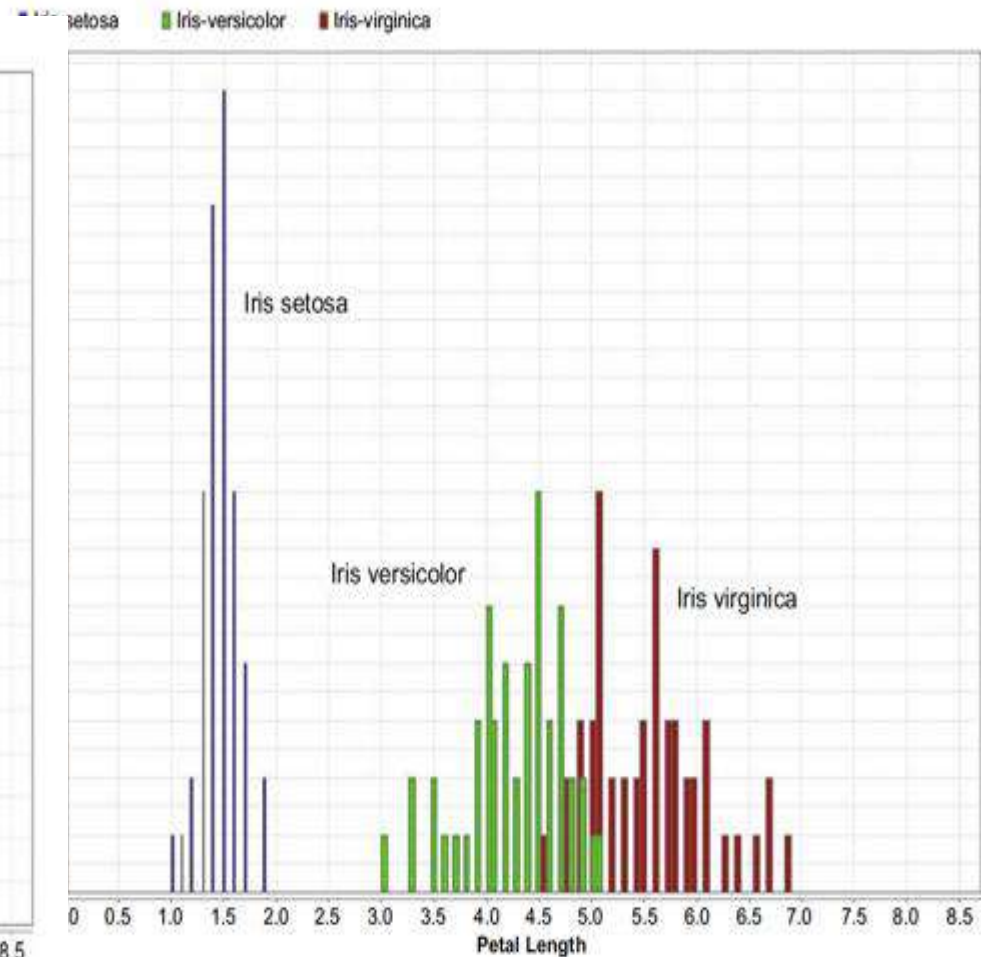
2.Relationships: Visualizing data in Cartesian coordinates enables exploration of the relationships between the attributes. Although representing more than three attributes on the x, y, and z-axes is not feasible in Cartesian coordinates, there are a few creative solutions available by changing properties like the size, color, and shape of data markers or using flow maps where more than two attributes are used in a two-dimensional medium.

Data Visualization-Univariate

A histogram is one of the most basic visualization techniques to understand the frequency of the occurrence of values. It shows the distribution of the data by plotting the frequency of occurrence in a range. Histograms are used to find the central location, range, and shape of distribution.



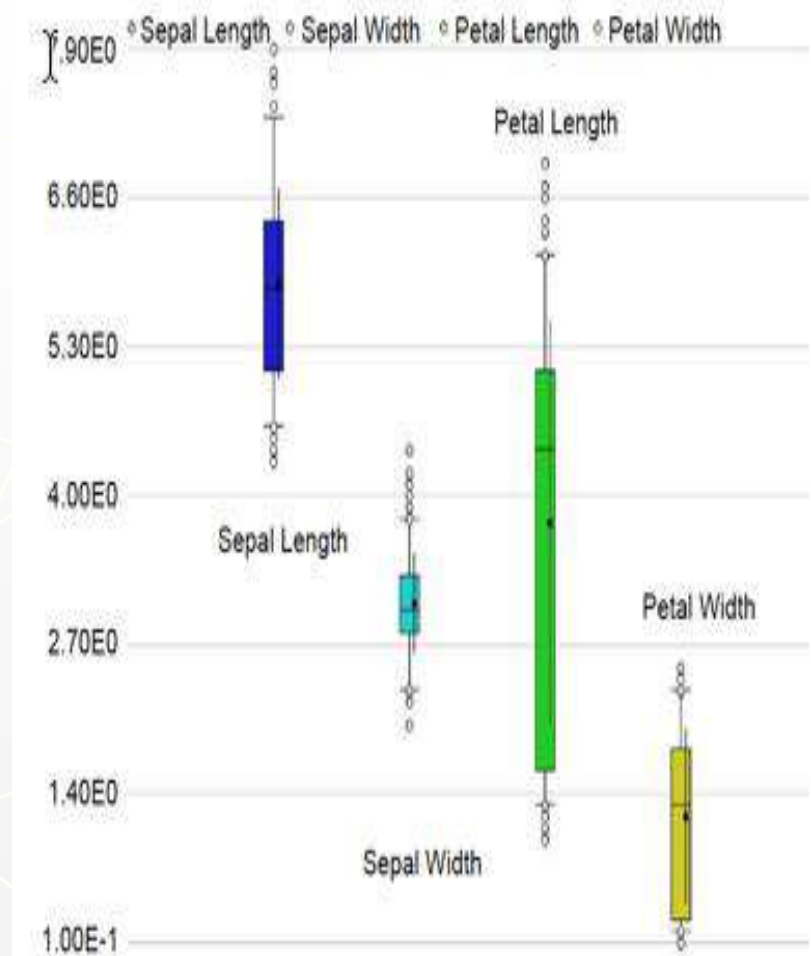
Histogram



Class stratified Histogram

Data Visualization

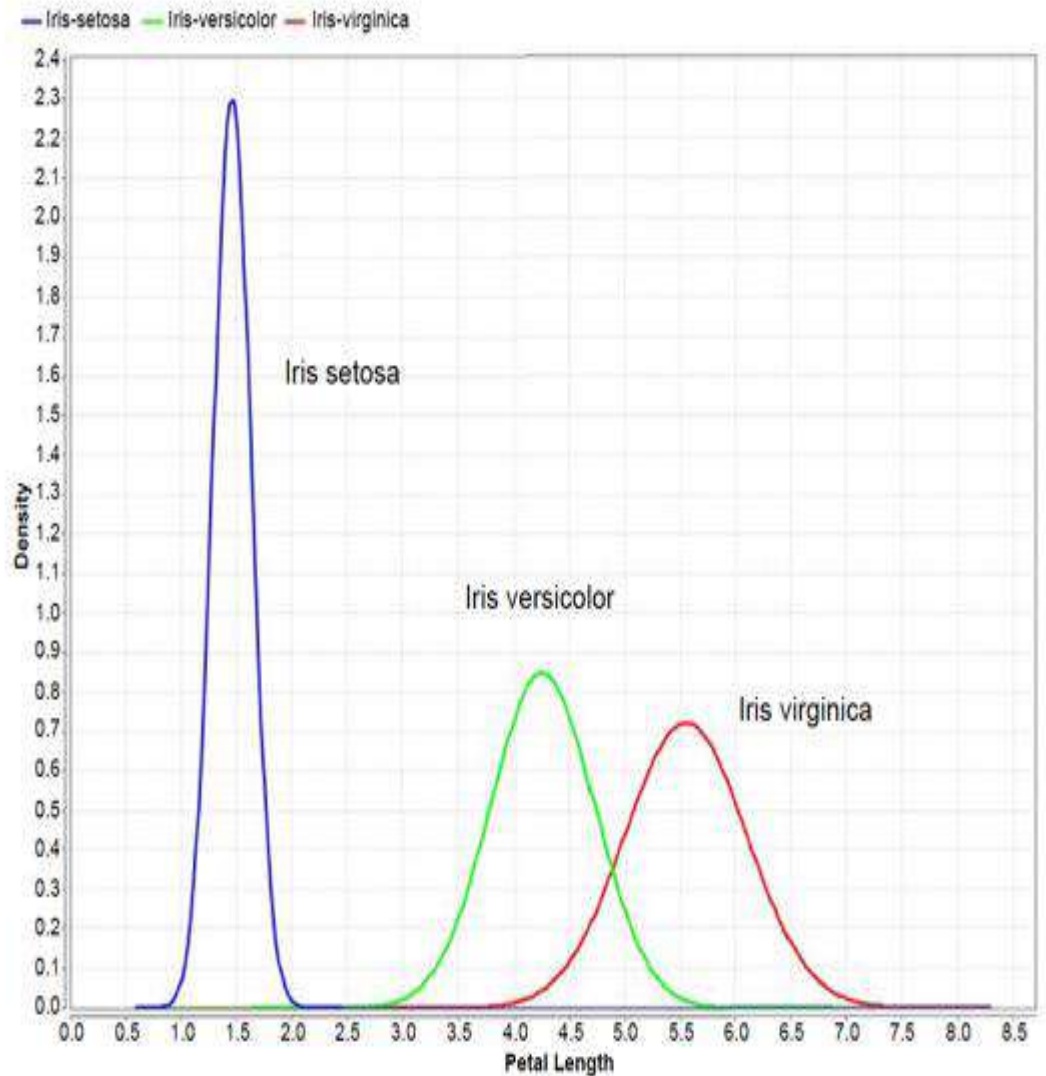
Box whisker plot is a simple visual way of showing the distribution of a continuous variable with information such as quartiles, median, and outlier overlaid by mean and standard deviation. The main attraction of box whisker or quartile charts is that distributions of multiple attributes can be compared side by side and the overlap between them can be deduced. The quartiles are denoted by Q1, Q2, and Q3 points, which indicate the data points with a 25% bin size. In a distribution, 25% of the data points will be below Q1, 50% will be below Q2, and 75% will be below Q3. The Q1 and Q3 points in a box whisker plot are denoted by the edges of the box. The Q2 point, the median of the distribution, is indicated by a cross line within the box



Quartile plot

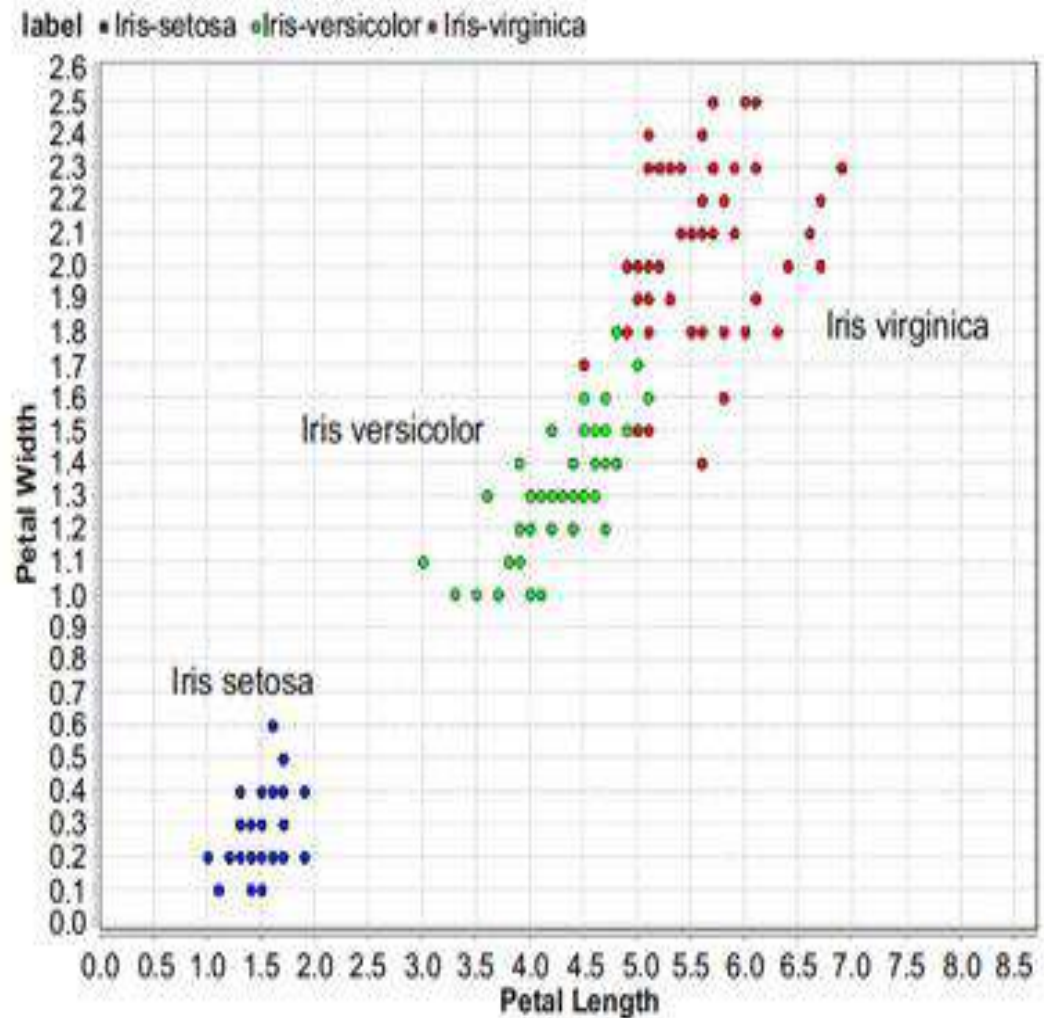
Data Visualization

Distribution plot



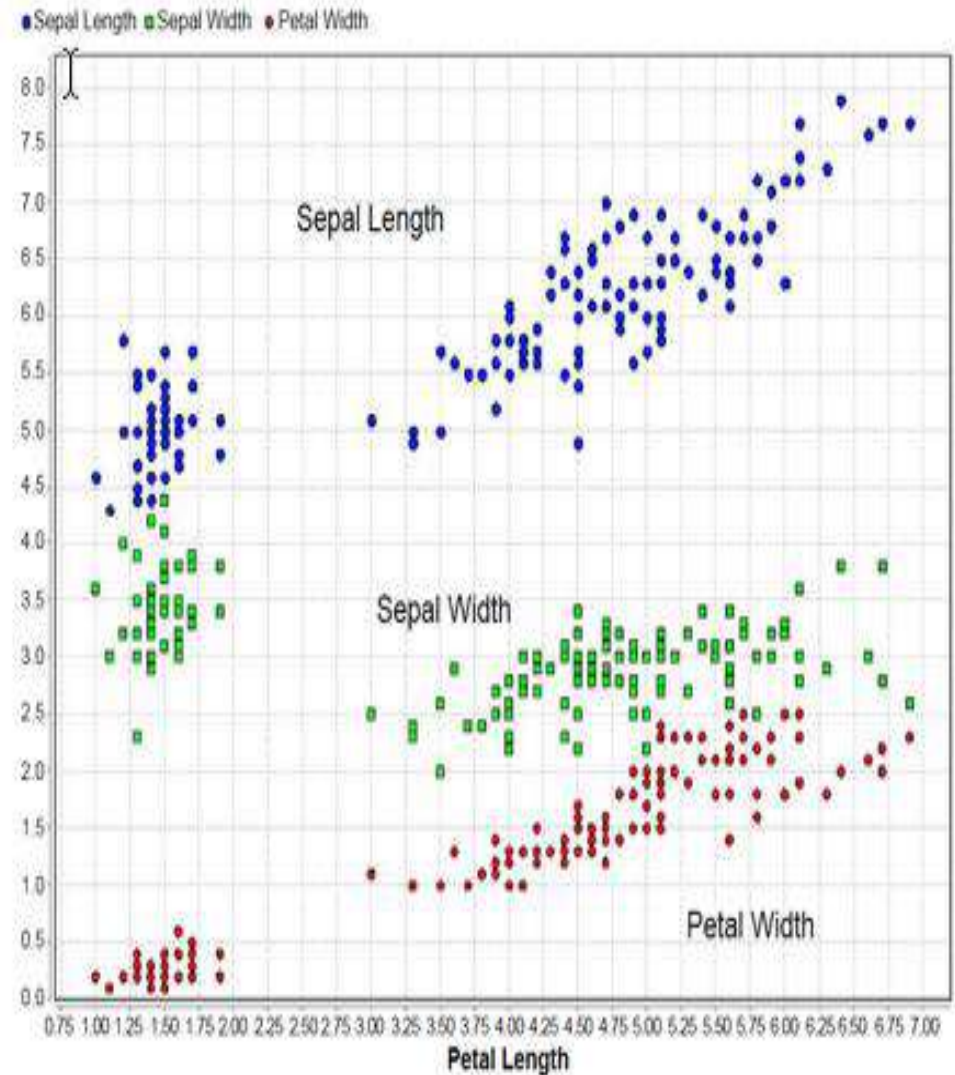
Data Visualization

Scatter plot



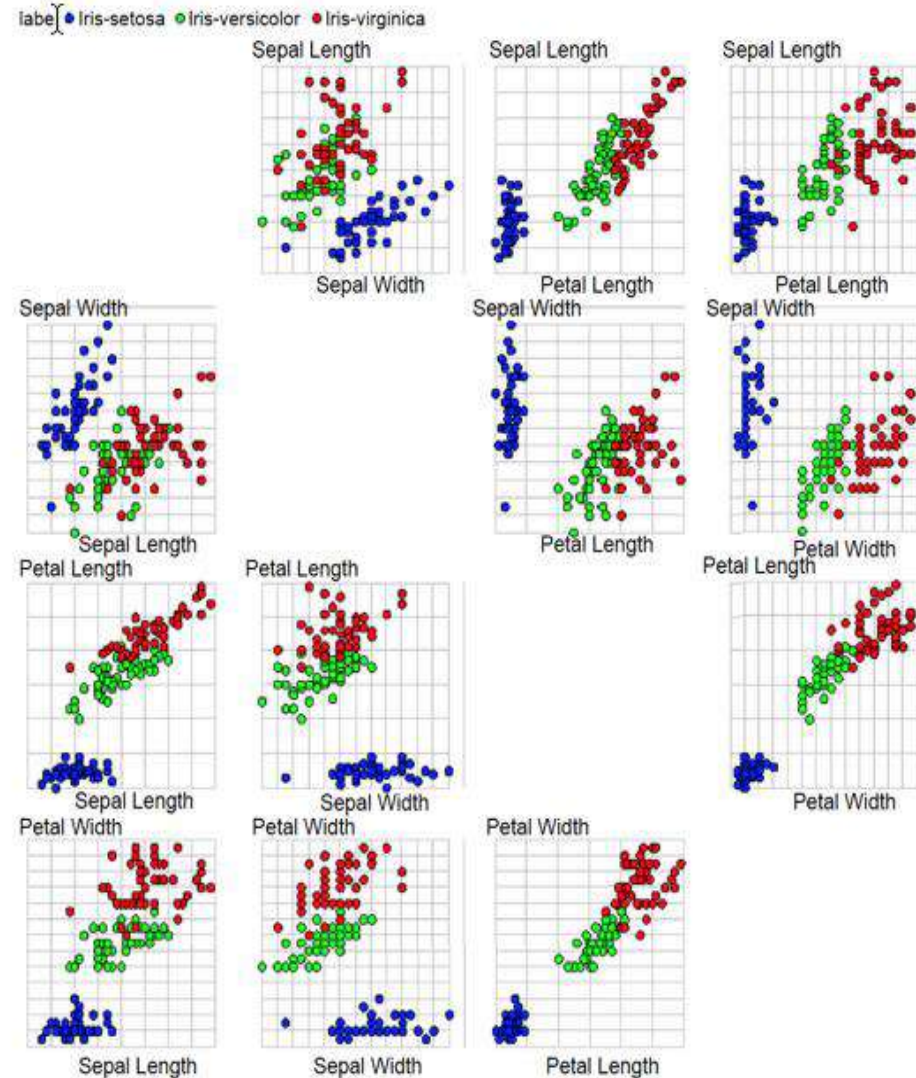
Data Visualization

Scatter mutiple



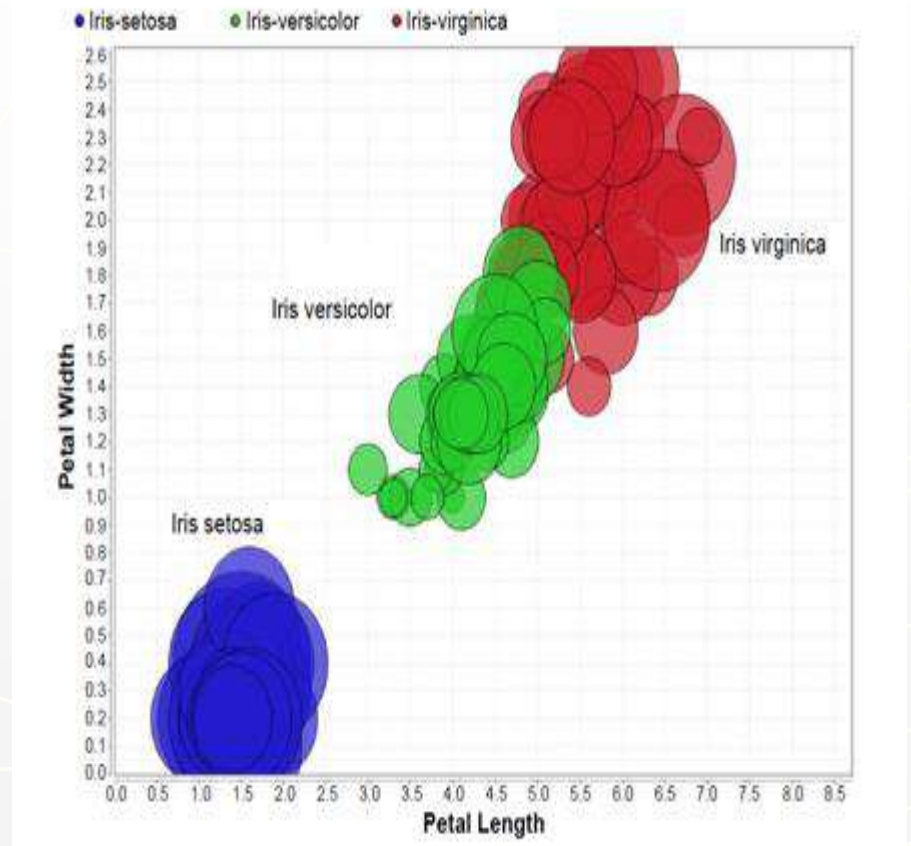
Data Visualization

Multiple Scatter matrix



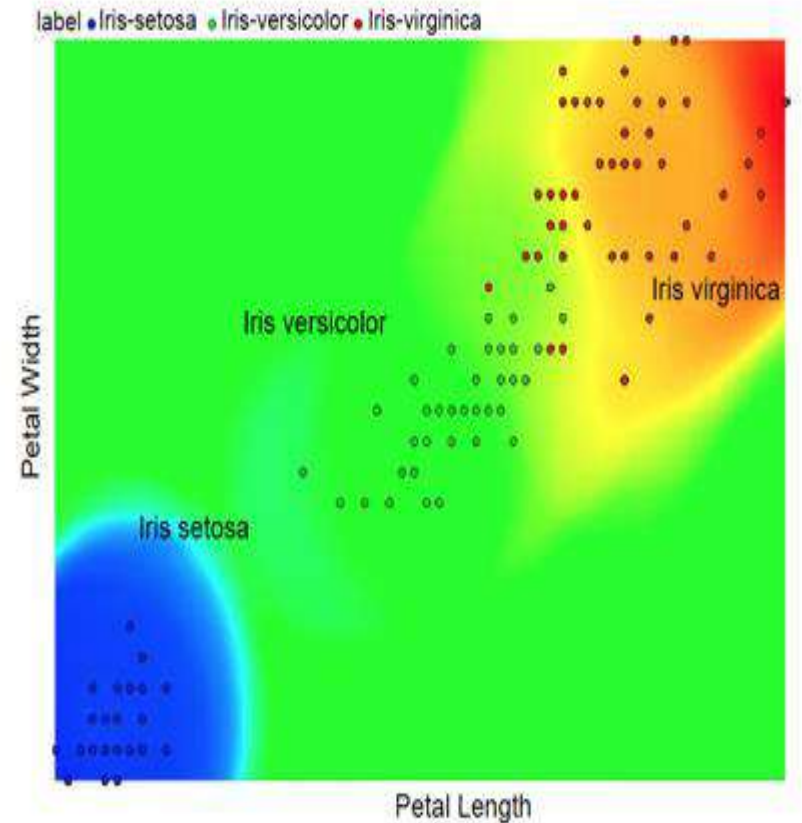
Data Visualization

Bubble plot



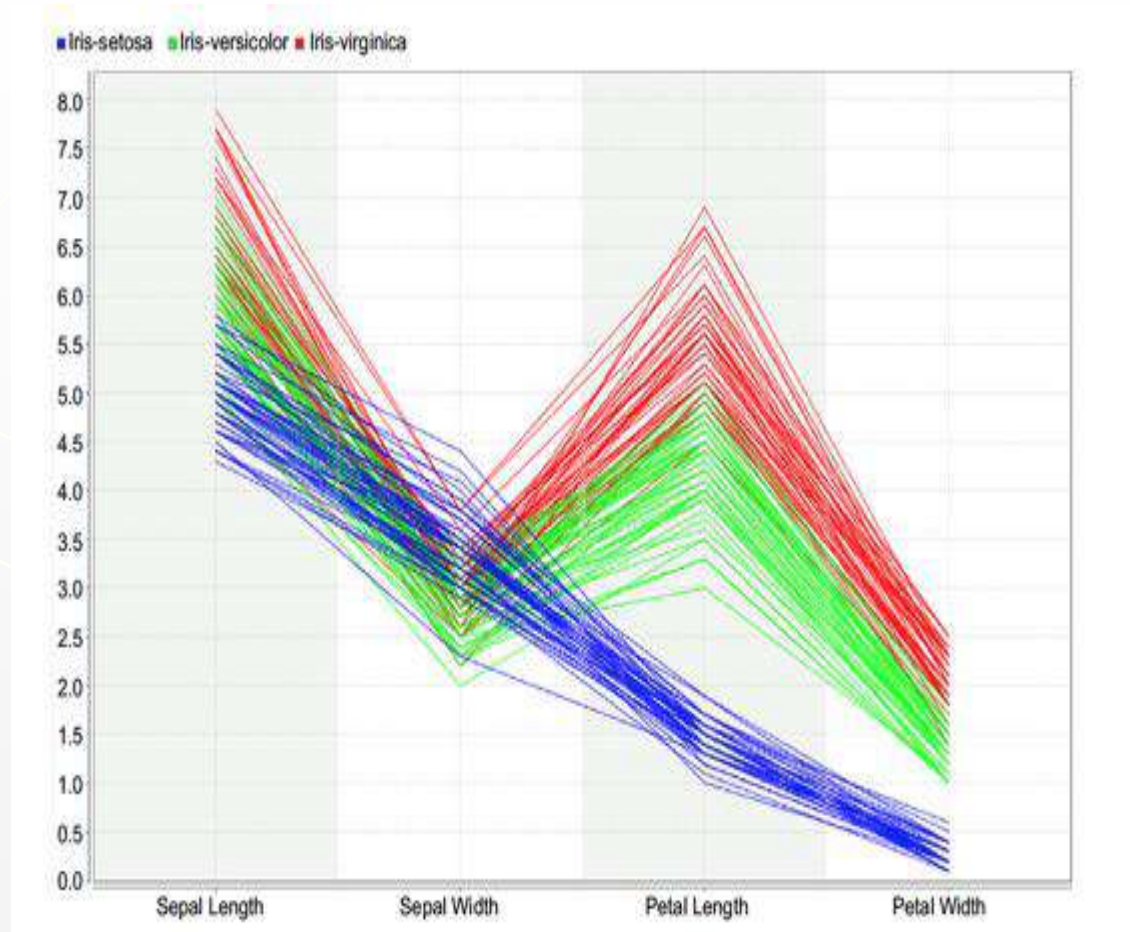
Data Visualization

Density chart



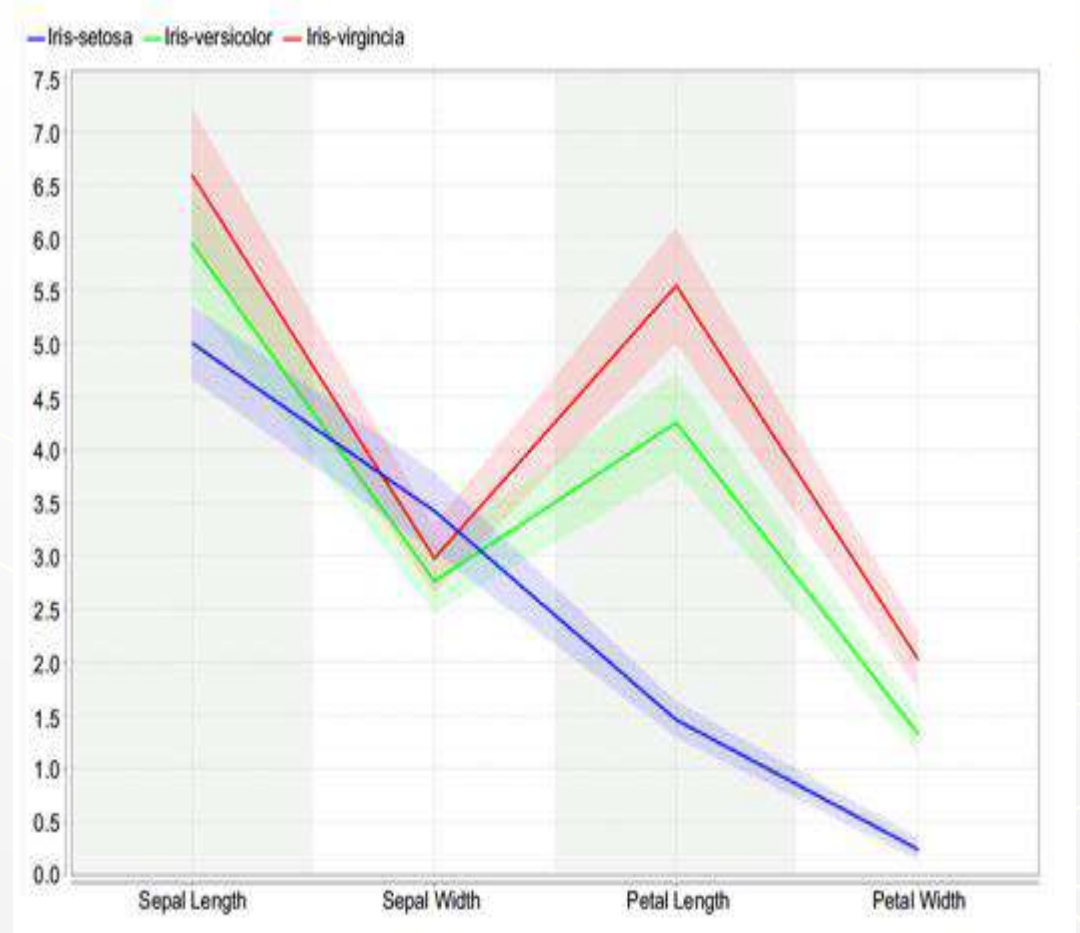
Data Visualization

Parallel chart



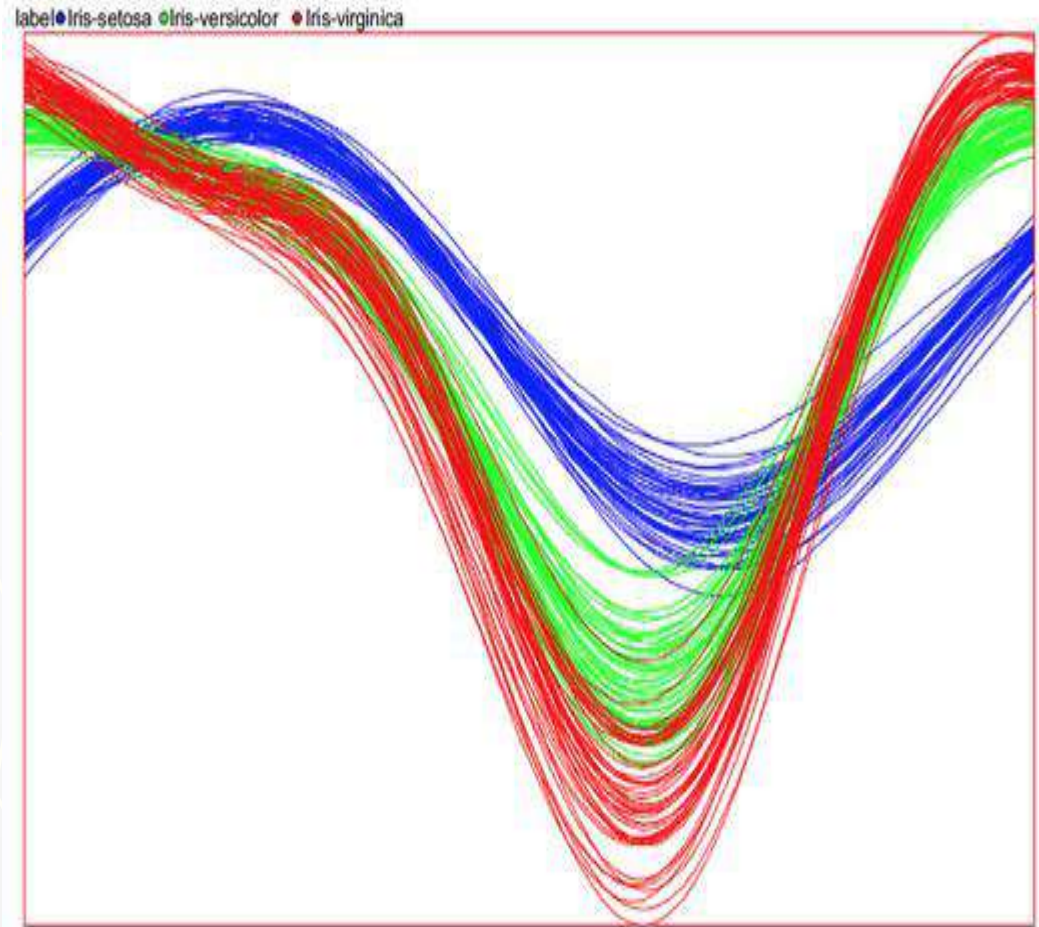
Data Visualization

Deviation chart



Data Visualization

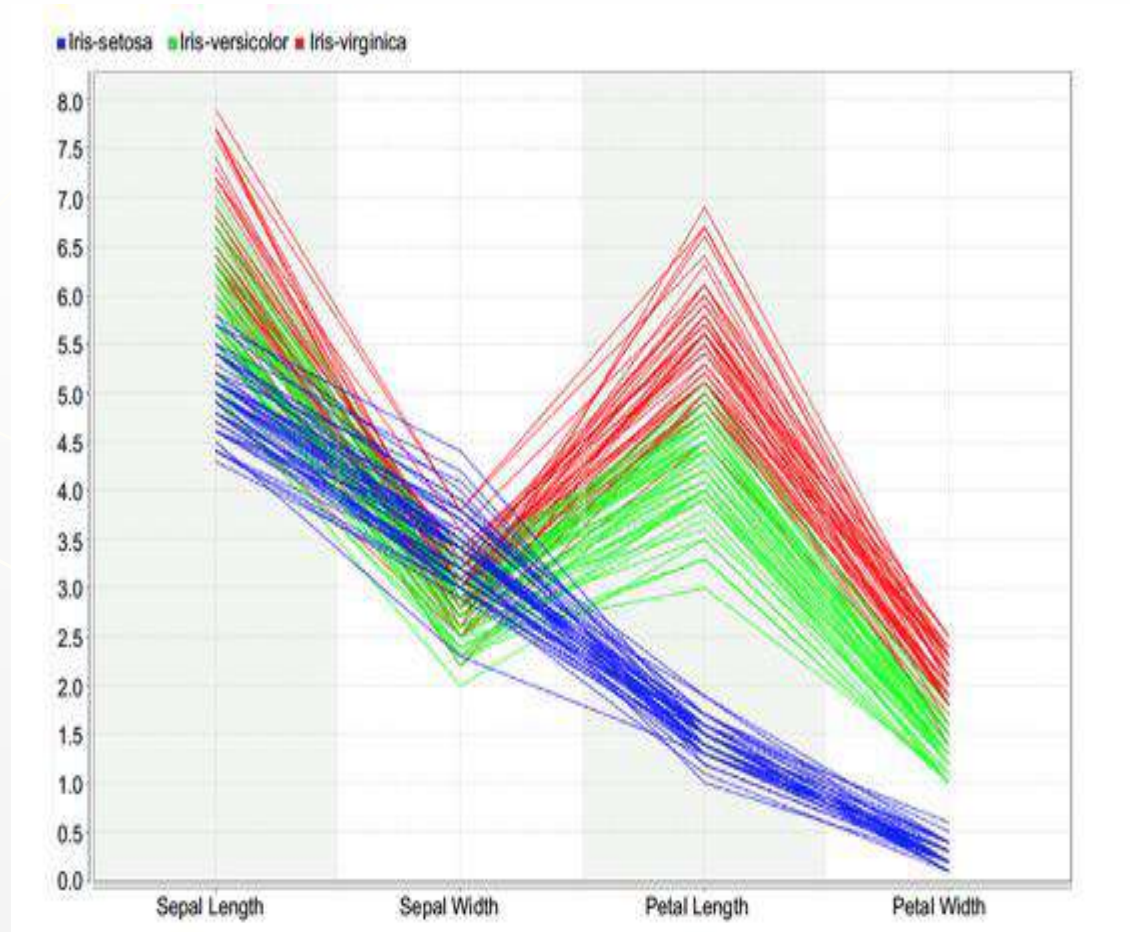
Andrews curves



$$f_x(t) = \frac{x_1}{\sqrt{2}} + x_2 \sin(t) + x_3 \cos(t) + x_4 \sin(2t) + x_5 \cos(2t) + \dots$$

Data Visualization

Parallel chart



Data Scraping

❑ What is Data Scraping?

Data scraping (also called web scraping) is the process of automatically collecting data from websites, documents, or other digital sources.

❑ Why is it needed?

Often, data is not available in ready-made datasets. We collect it from:

- Websites
- APIs
- Social media platforms
- Online databases
- PDFs, HTML pages, etc.

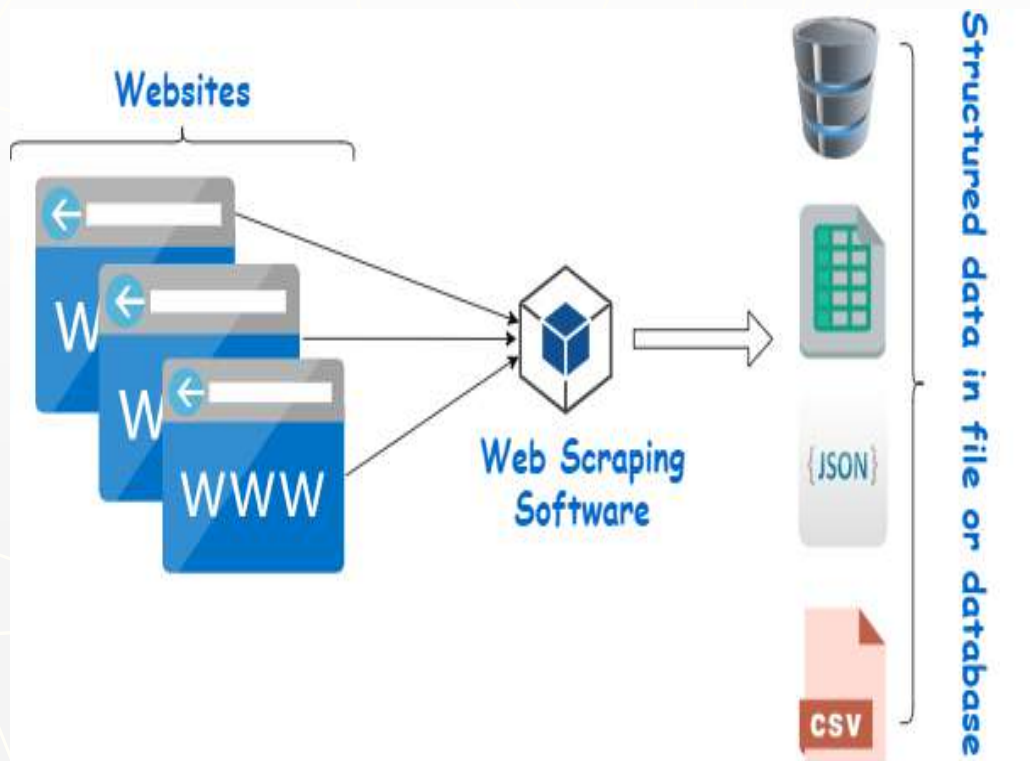
❑ Tools Used

- Python libraries:
 - BeautifulSoup
 - Scrapy
 - Selenium
 - requests
- APIs (Twitter API, Google Maps API, etc.)
- Web scraping frameworks

★ Example

Collecting:

- Product prices from an e-commerce website
- News headlines from a news portal
- Research data from public databases



Data Cleaning

What is Data Cleaning?

Data cleaning (data preprocessing) is the process of fixing or removing incorrect, incomplete, or inconsistent data.

- Raw data is usually:
- Messy
- Incomplete
- Noisy
- Inconsistent

★ Common Data Problems

- Missing values (NaN)
- Duplicate records
- Incorrect formats
- Outliers
- Typographical errors

★ Techniques Used

- Handling missing values:
 - ❖ Remove rows
 - ❖ Replace with mean/median/mode
- Removing duplicates
- Standardizing formats (date, currency, etc.)
- Outlier detection
- Data type conversion

★ Tools Used

- Python (Pandas, NumPy)
- R
- Excel
- SQL

Data Summarization

✦ What is Data Summarization?

Data summarization means presenting data in a compact and meaningful way to understand patterns and trends.

✦ Methods of Summarization

A) Statistical Summary

Mean, Median, Mode, Standard deviation, Variance, Min/Max

B) Tabular Summary

Grouping data, Pivot tables, Aggregation (sum, count, average)

C) Visual Summary

Bar charts, Histograms, Box plots, Pie charts, Line graphs

✦ Tools Used

Pandas (describe(), groupby()), Matplotlib / Seaborn, Tableau, Power BI

• Complete Workflow Example

Suppose you want to analyze online product reviews:

Scrape reviews from an e-commerce website.

Clean the data:

- Remove missing reviews
- Correct date formats
- Remove duplicates

Summarize:

- Average rating
- Most common words
- Monthly trend of ratings

Statistical Inference

Statistical Inference is the process of using sample data to make conclusions about a population.

1. Populations and Samples

Population

A population is the complete set of all individuals or items of interest in a study.

Examples:

- All students in a university
- All customers of a company
- All voters in a country

Population parameters include: μ (mean), σ^2 (variance), p (proportion)

Sample

A sample is a subset of the population selected for analysis.

Sample statistics include: $\bar{x}$ (mean), s^2 (variance), $\hat{p}$ (proportion)

Samples are used because collecting data from the entire population is often expensive and time-consuming.

Example

Suppose you want to know the average height of all students in a university.

Population → All students

Sample → 200 randomly selected students

Use sample mean ($\bar{x}$) to estimate population mean (μ)

2. Statistical Modeling

A statistical model is a mathematical representation of a real-world process using probability and equations. It helps us: Understand relationships, Make predictions, Test hypotheses

General form of a statistical model:

$$Y = f(X) + \varepsilon$$

Where:

- Y = Dependent variable
- X = Independent variable(s)
- ε = Random error term

Statistical Inference

Types of Statistical Models:

❑ Descriptive Models

Describe data patterns

Example: Mean, variance

❑ Predictive Models

Predict future outcomes

Example: Linear Regression

❑ Inferential Models

Make conclusions about population

Example: Hypothesis Testing

3. Probability Distributions

Probability distributions describe how values of a random variable are distributed.

Random Variable

A random variable is a numerical outcome of a random experiment.

Types:

- Discrete Random Variable (e.g., number of heads in coin toss)
- Continuous Random Variable (e.g., height, weight)

Common Probability Distributions

★ 1. Binomial Distribution (Discrete)

Used when:

- Fixed number of trials
- Two outcomes (success/failure)
- Constant probability

Example:

Tossing a coin 10 time

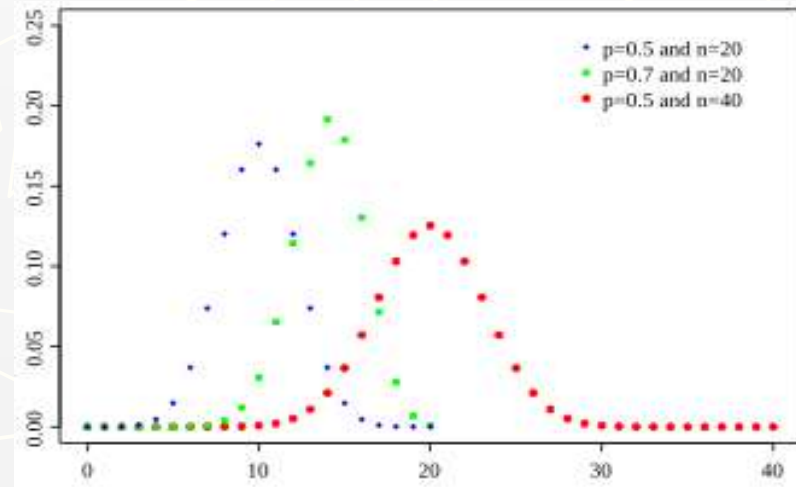
🎯 Simple Real-Life Example

Suppose a company wants to know:

“Is the average customer satisfaction above 4?”

Steps:

- Take a sample of 100 customers
- Calculate sample mean
- Assume normal distribution
- Perform hypothesis testing
- Draw conclusion about population



Statistical Inference

2. Poisson Distribution (Discrete)

Used when:

- Counting events
- Events occur independently
- Constant rate

Example:

Number of calls per hour

✦ 3. Normal Distribution (Continuous)

Most important distribution.

Properties:

Bell-shaped curve

Symmetrical

Mean = Median = Mode

Many real-world variables follow normal distribution: Heights, IQ scores, Measurement errors

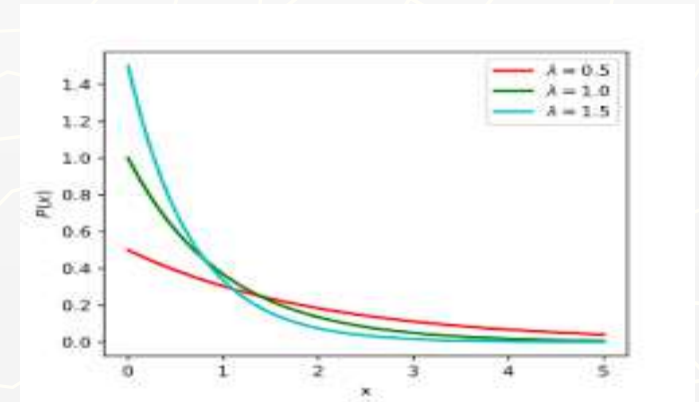
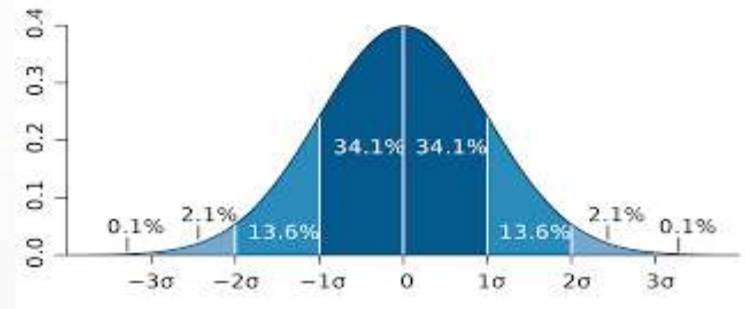
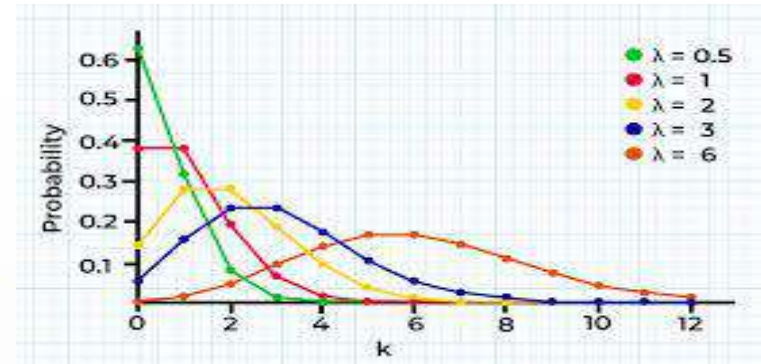
✦ 4. Exponential Distribution

Used for:

- Time between events
- Waiting time problems

Relationship Between These Concepts

- We collect a **sample**
- Build a **statistical model**
- Assume a **probability distribution**
- Estimate population parameters
- Perform hypothesis testing



Statistical Significance and P-values

What is Statistical Significance?

A result is statistically significant if it is unlikely to have occurred by random chance alone.

In simple terms:

If the result is very unlikely under the assumption of “no effect,” we say it is statistically significant.

Hypothesis Testing Framework

Statistical significance is determined through hypothesis testing.

Step 1: Define Hypotheses

Null Hypothesis (H_0) → No effect / No difference

Alternative Hypothesis (H_1) → There is an effect / Difference exists

Example:

H_0 : Average customer rating = 4

H_1 : Average customer rating $\neq$ 4

What is a P-value?

The p-value is the probability of obtaining results as extreme as the observed result, assuming the null hypothesis is true.

In simple words:

P-value tells us how strong the evidence is against the null hypothesis.

Decision Rule

We compare the p-value with a significance level (α).

Common significance levels:

$\alpha = 0.05$ (most common)

$\alpha = 0.01$

$\alpha = 0.10$

Rule:

If $p \leq \alpha$ → Reject H_0 (Statistically significant)

If $p > \alpha$ → Fail to reject H_0 (Not statistically significant)

Statistical Significance and P-values

Example

Suppose we test whether a new teaching method improves scores.

After analysis:

p-value = 0.03

$\alpha = 0.05$

Since:

$0.03 < 0.05 \rightarrow \text{Reject } H_0$

Conclusion:

The new teaching method has a statistically significant effect.

Type I and Type II Errors

Significance level (α) controls Type I error.

Error Type	Meaning
Type I Error	Rejecting a true H_0
Type II Error	Failing to reject a false H_0